

## ASME Journal of Fluid Engineering - Guest Editorial

### Special Issue: To Commemorate the Legacy of Malcolm J. Andrews - Scientist, Educator, and Leader



This special issue commemorates the scientific legacy of Dr. Malcolm J. Andrews, former Editor of the ASME Journal of Fluids Engineering (2010–2018). Malcolm devoted his career to the development and applications of computational, experimental, and analytical tools to investigate and solve diverse, complex fluid dynamics and heat transfer problems. Malcolm's most notable research accomplishments include: (i) contributions to innovative experiments, simulations, and theory to understand buoyancy-driven turbulent mixing due to Rayleigh–Taylor Instability; (ii) development of the multi-phase particle-in-cell (MP-PIC) method for understanding fragmentation in spray jets, coal-water slurry, and other dense particulate flows; (iii) mixing in nano- and bio-materials in applications ranging from protein glues to carbon nanotube composites; (iv) determining dispersion in proteins to improve therapeutic drug delivery via microencapsulation; (v) innovative methods for storing information using nano-pinned dipole magnetic vortices in superconducting materials; and, (vi) modeling and design modifications for shell-and-tube heat exchangers, as well as heat transfer in rotating systems such as turbine blades. The results of his research have been published in a long series of trend-setting journal and conference papers, book chapters, and patents.

Malcolm received his B.A. and M.A. in Mathematics from Oxford University (1980, 1988) and his Ph.D. and DIC in Mechanical Engineering from Imperial College London (1986). He moved to the United States after receiving his Ph.D. and first joined Princeton University as a postdoc (1986–

1991). Subsequently, Malcolm joined Texas A&M University as an Assistant Professor, where he rose to the rank of Full Professor (1991–2005). At Texas A&M, Malcolm served as the principal advisor to many Ph.D. and M.S. students. Malcolm was passionate about educating both inside and outside the classroom and mentoring the next generation of engineers, entrepreneurs, and thought-leaders in their respective fields. He taught numerous undergraduate and graduate courses in fluid mechanics and heat transfer and was instrumental in the development of an undergraduate Numerical Methods course. He received several awards in recognition of excellence in teaching and education at the undergraduate and graduate levels.

In 2005, Malcolm moved to Los Alamos National Laboratory (LANL) as one of the first National Security Fellows recruited to the U.S. Department of Energy National Laboratories. Soon after, he received the prestigious Ernest Orlando Lawrence Award (2006) for pioneering contributions in the area of fluid instabilities and turbulent mixing, with expertise spanning the realms of theory, numerical simulation, and experiment. Over the next decade (2005–2015), Malcolm was instrumental for the successful implementation of physics-based mix-models at LANL. He also played an active leadership role at Los Alamos, including as a Group Leader (XCP-4: Methods and Algorithms), Project Leader (Advanced Simulation and Computing Program Mix and Burn Project), and Founding Director of the CoMuEx (Center of Mixing under Extreme Conditions). Beginning in 2015, Malcolm served as an executive advisor from the Laboratory at the National Nuclear Security Administration (NNSA) headquarters in Washington, D.C., where he helped prepare the National Strategic Computing Initiative (NSCI) Program Plan. His brilliant career was cut short when he passed away on January 27, 2019, after a long and valiant battle with cancer.

Malcolm was also an active member of the ASME Fluids Engineering (FED) and Heat Transfer (HTD) Divisions for over three decades. He was a regular attendee of the ASME annual and summer meetings and served in various roles ranging from chairing sessions, organizing tracks and symposia, to being the co-organizer of the International Symposium on Numerical Methods for Multiphase Flows from 2000–2015. In addition, Malcolm served as Vice-Chair and Chair of the Multiphase Flow Technical Committee (MFTC) from 2006 to 2010. In recognition of his contributions to the fluids engineering community, Malcolm was designated a fellow of ASME in 2006. Malcolm started at the Journal of Fluids Engineering (JFE) in the role of an Associate Editor in 2003 before being appointed the Editor of the journal in 2010. During his tenure as an Editor, Malcolm led an effort, helped by the ASME Publications staff to correct several inconsistencies in reporting of ASME journal names in citations that ultimately led to the doubling of the impact factor of many of the ASME journals, including the JFE.

A celebration of Malcolm's life and scientific contributions was held during a special symposium at the ASME–JSME–KSME Joint Fluids Engineering Conference in San Francisco, CA (July 2019). This special issue was conceived after that conference by a group of Associate Editors of the JFE that included Malcolm's former Ph.D. students, collaborators, and colleagues. The Guest Editors made significant efforts to capture Malcolm's broad spectrum of research contributions in soliciting manuscripts for this Special Issue. The invited papers have been written by former colleagues and are devoted to a subset of the many topics that Malcolm investigated over the years. During his time at Texas A&M, Malcolm was instrumental in developing novel experiments

to study Rayleigh–Taylor instability. A review article on the experimental designs of Rayleigh–Taylor instability (Banerjee) and two research articles studying Rayleigh–Taylor (Morgan and Jacobs) and Richtmyer–Meshkov (Noble et al.) instabilities feature discussions of the state-of-the-art diagnostics and experimental designs in buoyancy- and shock-driven mixing. Throughout his career, Malcolm was instrumental in developing computational models for studying two-phase flows and heat transfer problems. To capture the many aspects of Malcom’s contributions, articles are included on: (a) implicit large-eddy simulations of Rayleigh–Taylor instability with periods of zero acceleration (Aslangil et al.); (b) comparison of dynamical aspects of Richtmyer–Meshkov instability (Zhou and Thornber); (c) the effect of shear on scalar transport in double-diffusive convection (Sichani et al.), and; (d) analysis of vortex structure development for pulsating flow past a square cylinder (Fowler et al.). In the last decade of his career at LANL, Malcolm was instrumental in leading a Laboratory Directed Research Development project “Turbulence by Design” for synergizing modeling, theory, and experiments to improve understanding of buoyancy- and shock-driven flows. To capture his many contributions, an elaborate review (Schilling) that highlights progress in Rayleigh–Taylor instability research using simulations, modeling, and experiments are included. In addition, three research articles to capture the breadth of Malcolm’s contributions to modeling complex fluid dynamics problems are included: (a) early-time modifications in buoyancy–drag models for Richtmyer–Meshkov instability (Youngs and Thornber); (b) comparison of various growth models for Rayleigh–Taylor and Richtmyer–Meshkov instabilities (Canfield et al.), and; (c) models of stress behavior in human blood due to oscillatory shear flow (Saengow et al.).

The editors are appreciative of all authors for their contributions and reviewers for their timely assistance in reviewing and improving the quality of the manuscripts. The Lead Guest Editor (Arindam Banerjee) would like to thank Francine Battaglia, Editor of the ASME Journal of Fluids Engineering, for allowing extensions as we navigated delays due to the COVID-19 pandemic. We are also thankful for the support from the JFE Editorial Office, Colette Montague, and the Production Team at the American Society of Mechanical Engineers for their work in producing this Special Issue.

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